



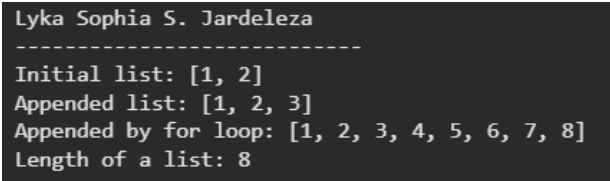
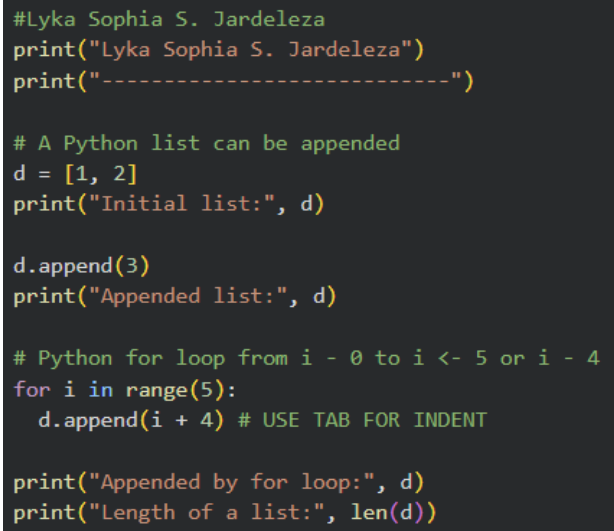
Name:	Lyka Sophia S. Jardeleza	Date:	01/07/2026
Program:	BSCpE	Section:	A461

A. Guided Exercises:

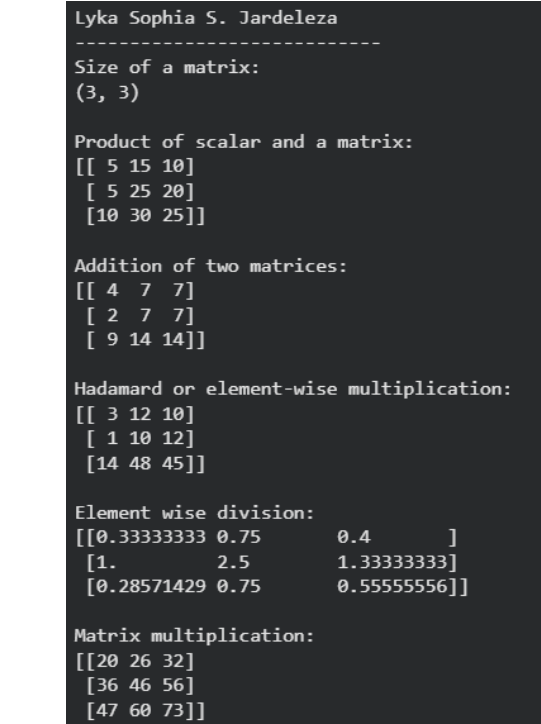
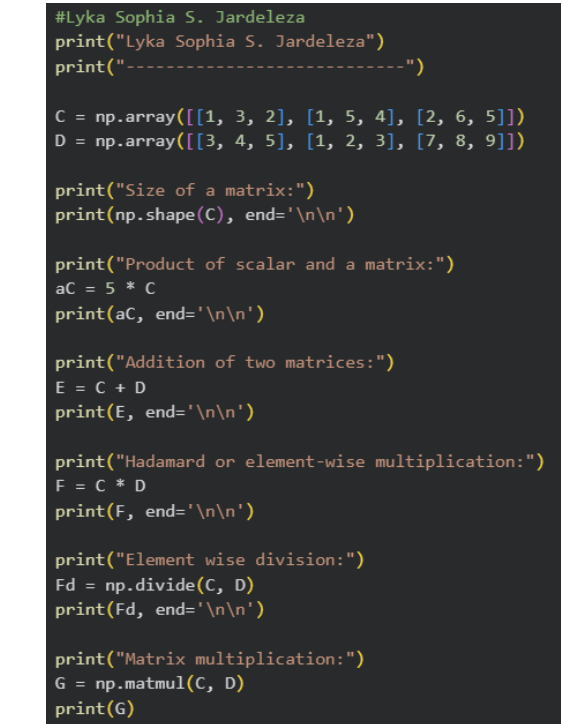
(Please include the following: 1. Photos of the codes and 2. Outputs after executing the Python code.)

- Choose five (5) from the guided exercises

1. Selected Exercise in Python List

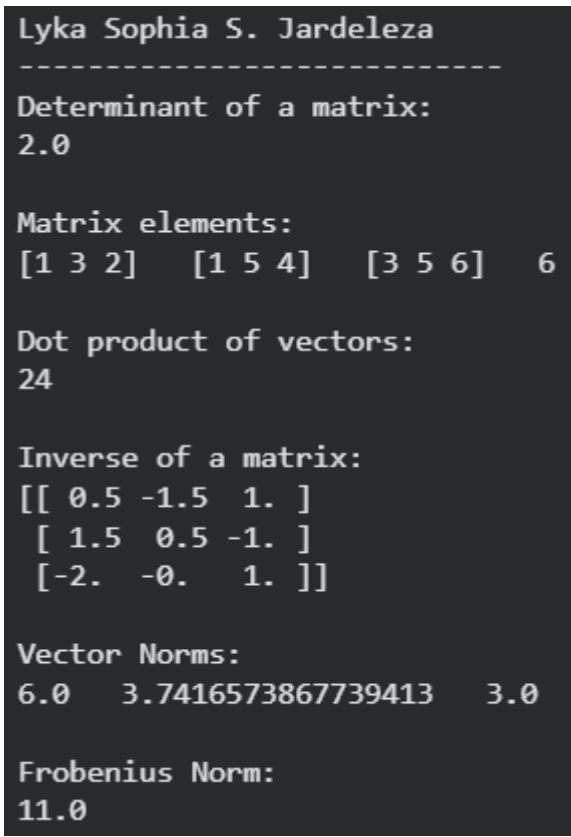
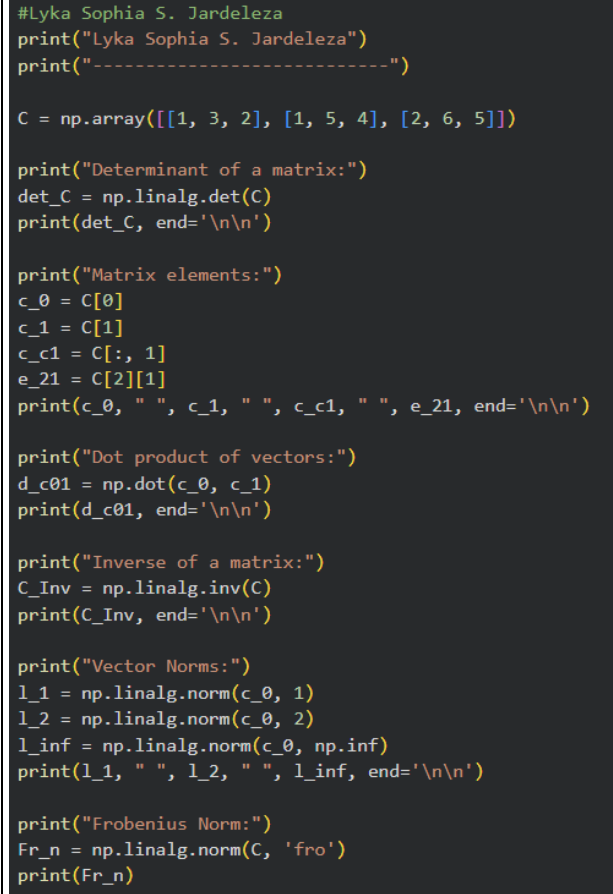
Photo/s of Python output	Photos of Python Codes in Colaboratory
 <pre>Lyka Sophia S. Jardeleza ----- Initial list: [1, 2] Appended list: [1, 2, 3] Appended by for loop: [1, 2, 3, 4, 5, 6, 7, 8] Length of a list: 8</pre>	 <pre>#Lyka Sophia S. Jardeleza print("Lyka Sophia S. Jardeleza") print("-----") # A Python list can be appended d = [1, 2] print("Initial list:", d) d.append(3) print("Appended list:", d) # Python for loop from i = 0 to i < 5 or i = 4 for i in range(5): d.append(i + 4) # USE TAB FOR INDENT print("Appended by for loop:", d) print("Length of a list:", len(d))</pre>

2. Selected Exercise in Basic Operations of Matrices

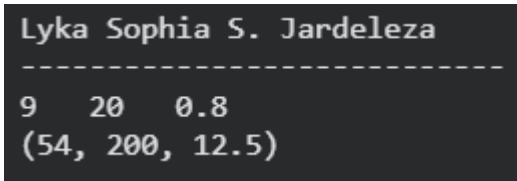
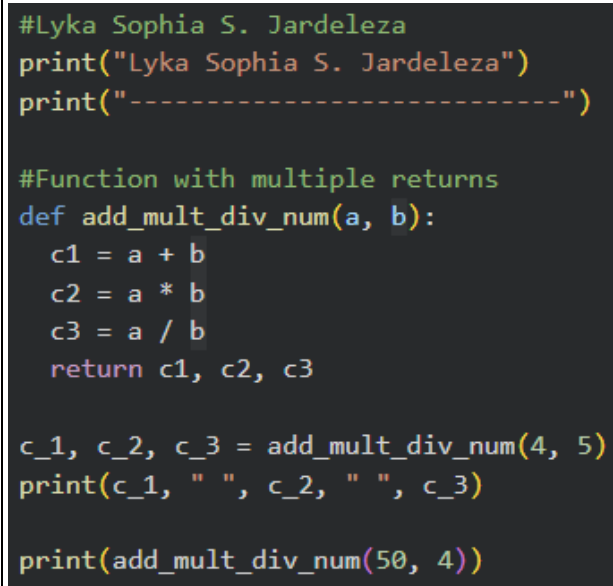
Photo/s of Python output	Photos of Python Codes in Colaboratory
 <pre>Lyka Sophia S. Jardeleza ----- Size of a matrix: (3, 3) Product of scalar and a matrix: [[5 15 10] [5 25 20] [10 30 25]] Addition of two matrices: [[4 7 7] [2 7 7] [9 14 14]] Hadamard or element-wise multiplication: [[3 12 10] [1 10 12] [14 48 45]] Element wise division: [[0.33333333 0.75 0.4] [1. 2.5 1.33333333] [0.28571429 0.75 0.55555556]] Matrix multiplication: [[20 26 32] [36 46 56] [47 60 73]]</pre>	 <pre>#Lyka Sophia S. Jardeleza print("Lyka Sophia S. Jardeleza") print("-----") C = np.array([[1, 3, 2], [1, 5, 4], [2, 6, 5]]) D = np.array([[3, 4, 5], [1, 2, 3], [7, 8, 9]]) print("Size of a matrix:") print(np.shape(C), end='\n\n') print("Product of scalar and a matrix:") aC = 5 * C print(aC, end='\n\n') print("Addition of two matrices:") E = C + D print(E, end='\n\n') print("Hadamard or element-wise multiplication:") F = C * D print(F, end='\n\n') print("Element wise division:") Fd = np.divide(C, D) print(Fd, end='\n\n') print("Matrix multiplication:") G = np.matmul(C, D) print(G)</pre>



3. Basic Linear Algebra Operations

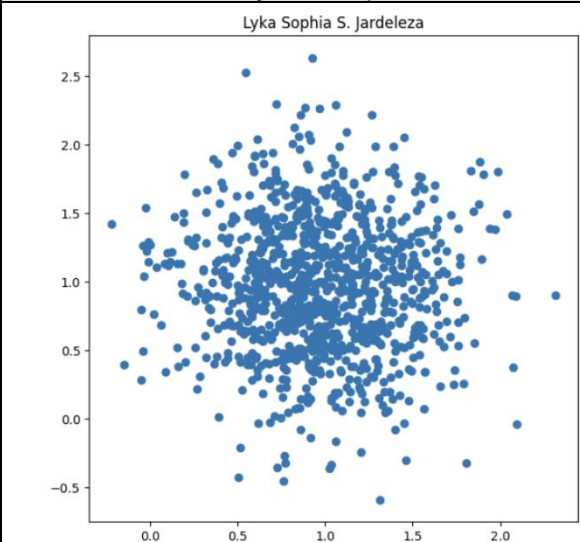
Photo/s of Python output	Photos of Python Codes in Colaboratory
	

4. Functions in Python

Photo/s of Python output	Photos of Python Codes in Colaboratory
	



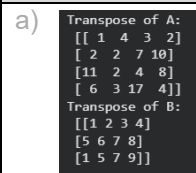
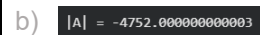
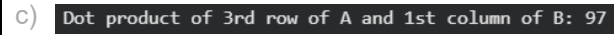
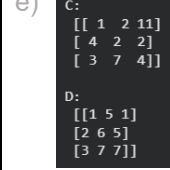
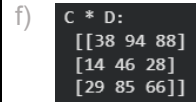
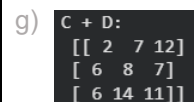
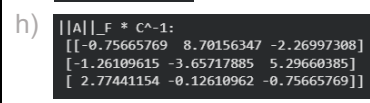
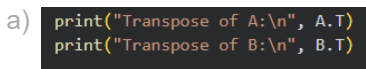
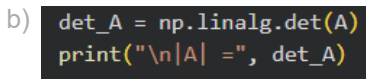
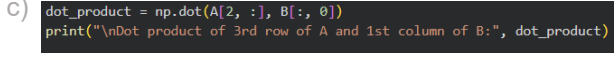
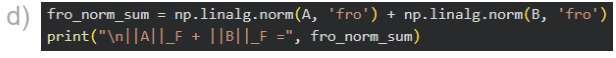
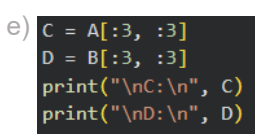
5. Selected Exercise in Matplotlib Plotting

Photo/s of Python output	Photos of Python Codes in Colaboratory
	<pre>#Lyka Sophia S. Jardeleza #Plotting a 2D Gaussian mu_x = 1 sigma_x = 0.4 x = np.random.normal(mu_x, sigma_x, 1000) mu_y = 1 sigma_y = 0.5 y = np.random.normal(mu_y, sigma_y, 1000) plt.figure(num=None, figsize=(7, 7)) plt.scatter(x, y) plt.title("Lyka Sophia S. Jardeleza") plt.show()</pre>

B. Problem Exercises:

(Please include the following: 1. Photos of the codes and 2. Outputs after executing the Python code.)

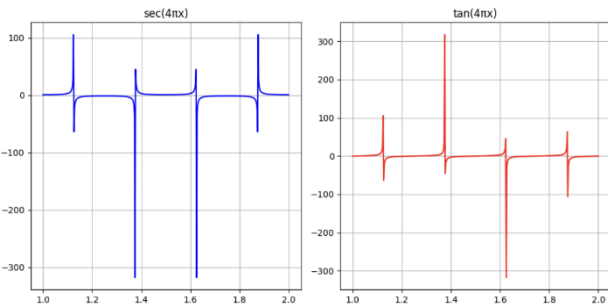
Problem 1:

Photo/s of Python output	Photos of Python Codes in Colaboratory
a)  b)  c)  d)  e)  f)  g)  h) 	<pre># Define matrices A = np.array([[1, 2, 11, 6], [4, 2, 2, 3], [3, 7, 4, 17], [2, 10, 8, 4]]) B = np.array([[1, 5, 1], [2, 6, 5], [3, 7, 7], [4, 8, 9]])</pre> a)  b)  c)  d)  e) 

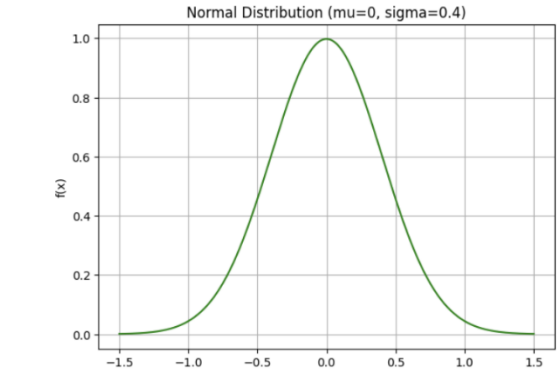


i) $B^T * A:$ <pre>[[26 67 59 79] [66 151 159 199] [60 151 121 176]]</pre> j) $D^{-1} _F + D = 12.534589286804266$	f) <code>CD = np.matmul(C, D)</code> <code>print("\nC * D:\n", CD)</code> g) <code>C_plus_D = C + D</code> <code>print("\nC + D:\n", C_plus_D)</code> h) <code>C_inv = np.linalg.inv(C)</code> <code>h_result = np.linalg.norm(A, 'fro') * C_inv</code> <code>print("\n A _F * C^-1:\n", h_result)</code> i) <code>B_TA = np.matmul(B.T, A)</code> <code>print("\nB^T * A:\n", B_TA)</code> j) <code>D_inv = np.linalg.inv(D)</code> <code>result_j = np.linalg.norm(D_inv, 'fro') + np.linalg.det(D)</code> <code>print("\n D^-1 _F + D =", result_j)</code>
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Problem 2:

Photo/s of Python output	Photos of Python Codes in Colaboratory
	<pre>import matplotlib.pyplot as plt import numpy as np # x values x = np.linspace(1.0, 2.0, 500) # functions sec = 1 / np.cos(4 * np.pi * x) tan = np.tan(4 * np.pi * x) # Subplots plt.figure(figsize=(10, 5)) plt.subplot(1, 2, 1) plt.plot(x, sec, 'b') plt.title("sec(4πx)") plt.grid(True) plt.subplot(1, 2, 2) plt.plot(x, tan, 'r') plt.title("tan(4πx)") plt.grid(True) plt.tight_layout() plt.show()</pre>

Problem 3:

Photo/s of Python output	Photos of Python Codes in Colaboratory
	<pre>import matplotlib.pyplot as plt import numpy as np # Normal distribution parameters mu = 0.0 sigma = 0.4 x = np.linspace(-1.5, 1.5, 500) f_x = (1/(sigma*np.sqrt(2*np.pi))) * np.exp(-0.5 * ((x-mu)/sigma)**2) plt.figure(figsize=(7,5)) plt.plot(x, f_x, 'g') plt.title("Normal Distribution (mu=0, sigma=0.4)") plt.xlabel("x") plt.ylabel("f(x)") plt.grid(True) plt.show()</pre>



Problem 4:

Photo/s of Python output	Photos of Python Codes in Colaboratory
a) <pre> mult_table(13): [[1 2 3 4 5 6 7 8 9 10 11 12 13] [2 4 6 8 10 12 14 16 18 20 22 24 26] [3 6 9 12 15 18 21 24 27 30 33 36 39] [4 8 12 16 20 24 28 32 36 40 44 48 52] [5 10 15 20 25 30 35 40 45 50 55 60 65] [6 12 18 24 30 36 42 48 54 60 66 72 78] [7 14 21 28 35 42 49 56 63 70 77 84 91] [8 16 24 32 40 48 56 64 72 80 88 96 104] [9 18 27 36 45 54 63 72 81 90 99 108 117] [10 20 30 40 50 60 70 80 90 100 110 120 130] [11 22 33 44 55 66 77 88 99 110 121 132 143] [12 24 36 48 60 72 84 96 108 120 132 144 156] [13 26 39 52 65 78 91 104 117 130 143 156 169]] </pre>	<pre> import numpy as np def mult_table(n): # Create multiplication table from 1 to n table = np.outer(np.arange(1, n+1), np.arange(1, n+1)) return table </pre> a) <pre> print("mult_table(13):\n", mult_table(13)) </pre> b) <pre> print("\nmult_table(25):\n", mult_table(25)) </pre>
b) <pre> mult_table(25): [[1 2 3 4 5 6 7 8 9 10 11 12 13 14 15 16 17 18 19 20 21 22 23 24 25] [2 4 6 8 10 12 14 16 18 20 22 24 26 28 30 32 34 36 38 40 42 44 46 48 50] [3 6 9 12 15 18 21 24 27 30 33 36 39 42 45 48 51 54 57 60 63 66 69 72 75] [4 8 12 16 20 24 28 32 36 40 44 48 52 56 60 64 68 72 76 80 84 88 92 96 100] [5 10 15 20 25 30 35 40 45 50 55 60 65 70 75 80 85 90 95 100 105 110 115 120 125] [6 12 18 24 30 36 42 48 54 60 66 72 78 84 90 96 102 108 114 120 126 132 138 144 150] [7 14 21 28 35 42 49 56 63 70 77 84 91 98 105 112 119 126 133 140 147 154 161 168 175] [8 16 24 32 40 48 56 64 72 80 88 96 104 112 120 128 136 144 152 160 168 176 184 192 200] [9 18 27 36 45 54 63 72 81 90 99 108 117 126 135 144 153 162 171 180 189 198 207 216 225] [10 20 30 40 50 60 70 80 90 100 110 120 130 140 150 160 170 180 190 200 210 220 230 240 250] [11 22 33 44 55 66 77 88 99 110 121 132 143 154 165 176 187 198 209 220 231 242 253 264 275] [12 24 36 48 60 72 84 96 108 120 132 144 156 168 180 192 204 216 228 240 252 264 276 288 300] [13 26 39 52 65 78 91 104 117 130 143 156 169 182 195 208 221 234 247 260 273 286 299 312 325] [14 28 42 56 70 84 98 112 126 140 154 168 182 196 210 224 238 252 266 280 294 308 322 336 350] [15 30 45 60 75 90 105 120 135 150 165 180 195 210 225 240 255 270 285 300 315 330 345 360 375] [16 32 48 64 80 96 112 128 144 160 176 192 208 224 240 256 272 288 304 320 336 352 368 384 400] [17 34 51 68 85 102 119 136 153 170 187 204 221 238 255 272 289 306 323 340 357 374 391 408 425] [18 36 54 72 90 108 126 144 162 180 198 216 234 252 270 288 306 324 342 360 378 396 414 432 450] [19 38 57 76 95 114 133 152 171 190 209 228 247 266 285 304 323 342 361 380 399 418 437 456 475] [20 40 60 80 100 120 140 160 180 200 220 240 260 280 300 320 340 360 380 400 420 440 460 480 500] [21 42 63 84 105 126 147 168 189 210 231 252 273 294 315 336 357 378 399 420 441 462 483 504 525] [22 44 66 88 110 132 154 176 198 220 242 264 286 308 330 352 374 396 418 440 462 484 506 528 550] [23 46 69 92 115 138 161 184 207 230 253 276 299 322 345 368 391 414 437 460 483 506 529 552 575] [24 48 72 96 120 144 168 192 216 240 264 288 312 336 360 384 408 432 456 480 504 528 552 576 600] [25 50 75 100 125 150 175 200 225 250 275 300 325 350 375 400 425 450 475 500 525 550 575 600 625]] </pre>	



C. Findings ,Observations, and Comments:

Guiding Questions :

1. *What did you learn in the guided exercises? Did you learn something new?*
2. *What are your findings, observations, and comments in the guided exercises?*
3. *What are your findings, observations, and comments in the problem exercises?*
4. *Did you experience any difficulties in solving the problem exercises?*

In the guided exercises, I learned the basic use of Python for simple programming tasks. I learned how to write basic codes, use lists, work with numbers, and apply simple functions. I also learned how NumPy helps in handling matrices and basic math operations faster and easier. Using Matplotlib was new to me, especially creating simple graphs to show data visually. Overall, I learned new skills that helped me understand how Python can be used for data and math-related tasks.

My findings and observations in the guided exercises showed that following step-by-step instructions made learning easier. The examples helped me see how the code works and how changes in the code affect the output. I observed that Python is easy to read and understand compared to other languages. The guided exercises helped build confidence before moving on to harder problems.

In the problem exercises, I found that applying what I learned was more challenging but helpful. These exercises required more thinking because there were fewer instructions. I observed that understanding the problem first made coding easier. The problem exercises helped me practice logical thinking and improve my ability to write correct code on my own.

I experienced some difficulties in solving the problem exercises, especially when I made small errors in typing or forgot certain steps. Some outputs were not correct at first, which required me to review and fix my code. However, these difficulties helped me learn from my mistakes and understand the concepts better.